

Robustness or Weighting? Re-Evaluating Federated Aggregation on Heterogeneous Health Data

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Motivation & Questions

- Hospitals and health agencies hold records that could sharpen how we predict disease, but privacy law stops them pooling the data.
- Federated learning trains one shared model without moving any patient record out of the institution holding it.
- The rule that merges each site's work is normally chosen for its robustness to noisy or unusual sites.
- Robust rules are scored against FedAvg, which weights each site by how much data it holds -- and they discard that weighting entirely, so two things change at once and nobody had separated them.
- We ask: when a robust rule wins, is it the robust statistic winning, or just the end of the loudest vote?

Background: how federated learning works

- Each site trains on its own data and sends only the model update -- no patient record ever leaves the institution holding it.

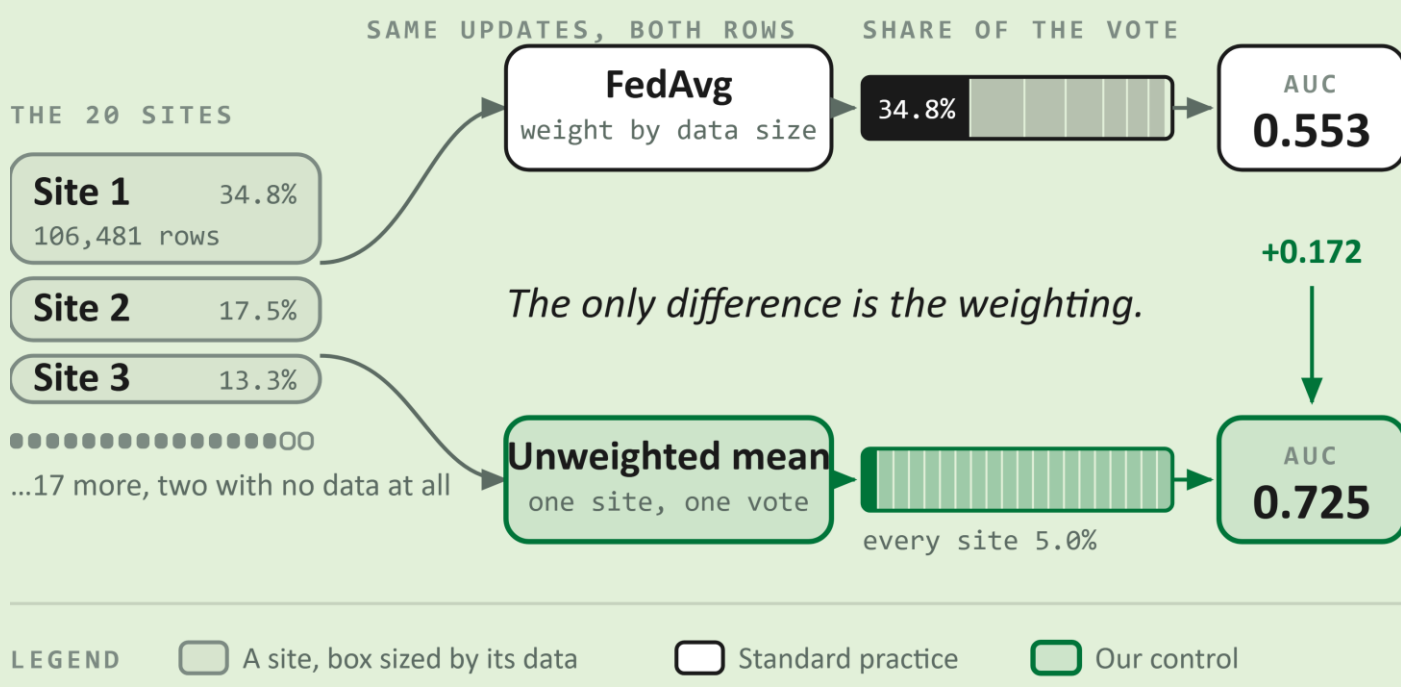


Figure 1. One training round, run two ways. The same 20 sites send the same updates; only the merge rule differs. FedAvg hands the largest site 34.8% of the vote, the control gives every site 5% -- worth +0.172 AUC on its own.

Dataset: two health datasets, split 20 ways

- CDC BRFSS 2023: 382,709 U.S. adults, predicting self-reported depression, which 20.7% of respondents report.
- UCI Breast Cancer Wisconsin runs alongside as a clean clinical benchmark.
- Each dataset is dealt to 20 simulated sites by Dirichlet partitioning, whose dial alpha sets how lopsided the split is.

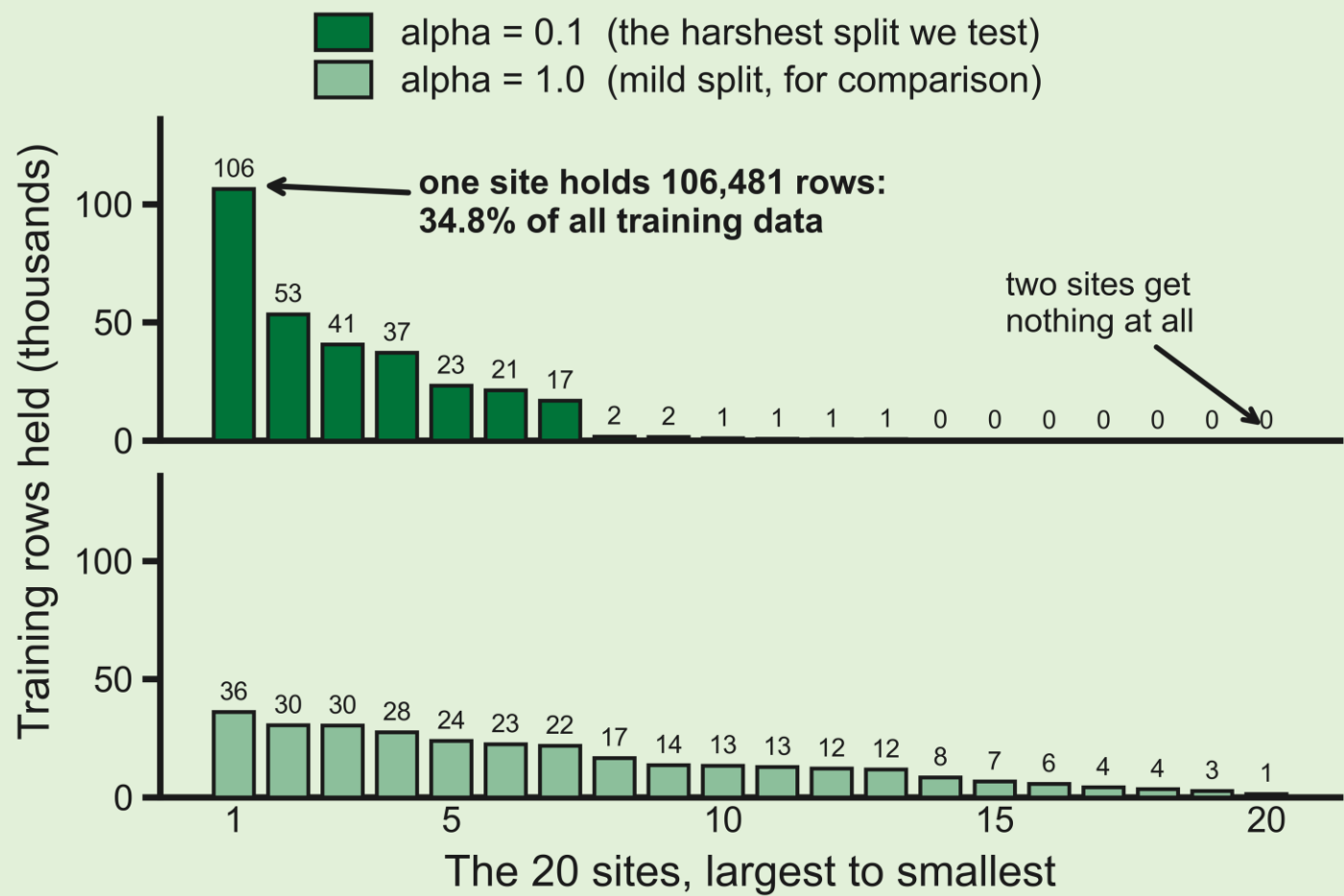


Figure 2. How the training data is dealt across 20 sites. At alpha = 0.1 one site holds 34.8% of it and two sites get none; at alpha = 1.0 the split is mild. This skew is what gives FedAvg's size-weighting something to get wrong.

References

[1] McMahan et al. Communication-efficient learning of deep networks from decentralized data. AISTATS, 2017.

[2] Blanchard et al. Machine learning with adversaries: Byzantine tolerant gradient descent. NIPS, 2017.

[3] Yin et al. Byzantine-robust distributed learning: towards optimal statistical rates. ICML, 2018.

[4] Karimireddy et al. Byzantine-robust learning on heterogeneous datasets via bucketing. ICLR, 2022.

[5] Hsu et al. Measuring the effects of non-identical data distribution for federated visual classification. arXiv:1909.06335, 2019.

[6] CDC. BRFSS 2023 survey data and documentation.

Methods: seven merge rules and one control

- Seven merge rules run head to head: FedAvg, FedProx, CS-Agg, Krum, trimmed mean, coordinate-wise median, and an unweighted mean.
- The unweighted mean is our control: it uses no robust statistic and differs from FedAvg only in ignoring how much data each site holds.
- Anything the control gains over FedAvg is therefore weighting alone, which is what isolates the effect.
- On BRFSS we corrupt labels only the way real people err, reporting no depression when they have it.

What we varied	Levels tested
Merge rules	7, including the unweighted-mean control
Datasets	BRFSS 2023 and Breast Cancer Wisconsin
Split skew, alpha	0.1, 0.5, 1.0 — lower is more lopsided
Label noise	0, 10, 20, 30% of records, at 20% of sites
Every run	20 sites, 50 rounds, seeds 42 / 123 / 456
Total	624 completed runs

Results

Most of the gain over FedAvg is weighting, not robustness

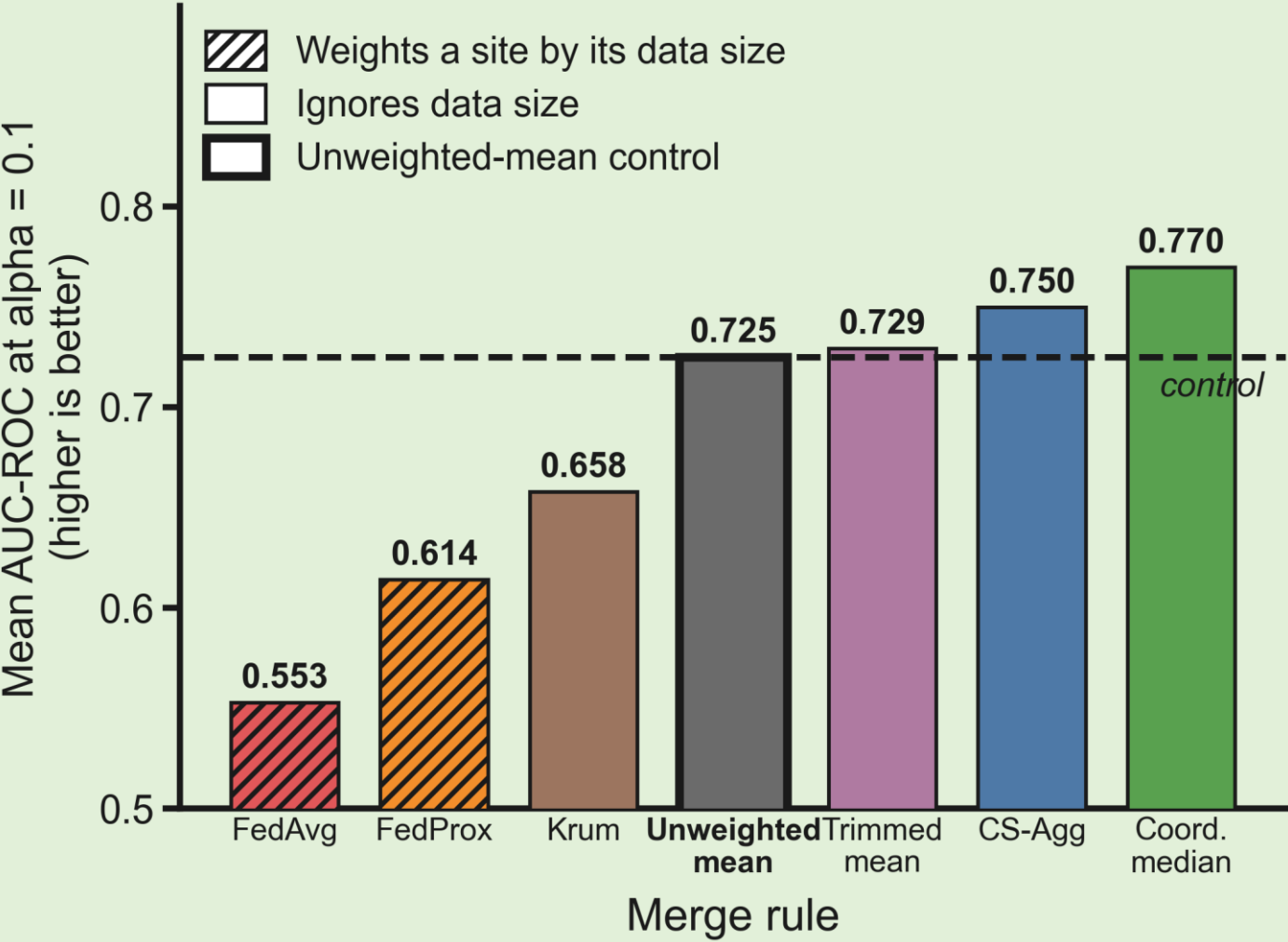


Figure 3. All seven rules under the harshest split, 12 runs each. Three robust rules clear the dashed control line, but only by 0.004 to 0.045, against the 0.172 the control alone gains over FedAvg.

Merge rule	Mean AUC	Worst run	vs. FedAvg	vs. control	Holm p vs. control
FedAvg	0.7081	0.4002	—	-0.0584	0.0036
FedProx	0.7286	0.4874	+0.0205	-0.0379	0.0036
Krum	0.7439	0.3275	+0.0358	-0.0226	0.22 n.s.
Unweighted mean	0.7665	0.6405	+0.0584	—	control
Trimmed mean	0.7691	0.6143	+0.0610	+0.0026	0.0036
CS-Agg	0.7746	0.7025	+0.0665	+0.0081	0.22 n.s.
Coord. median	0.7828	0.7279	+0.0747	+0.0163	< 0.0001

36 runs per rule on BRFSS, every condition pooled. Holm-corrected Wilcoxon signed-rank against the control.

- Pooled across every condition the control recovers 78 to 96% of what the three strongest robust rules gain, and CS-Agg cannot be told apart from it.

Krum is unreliable, not merely weaker

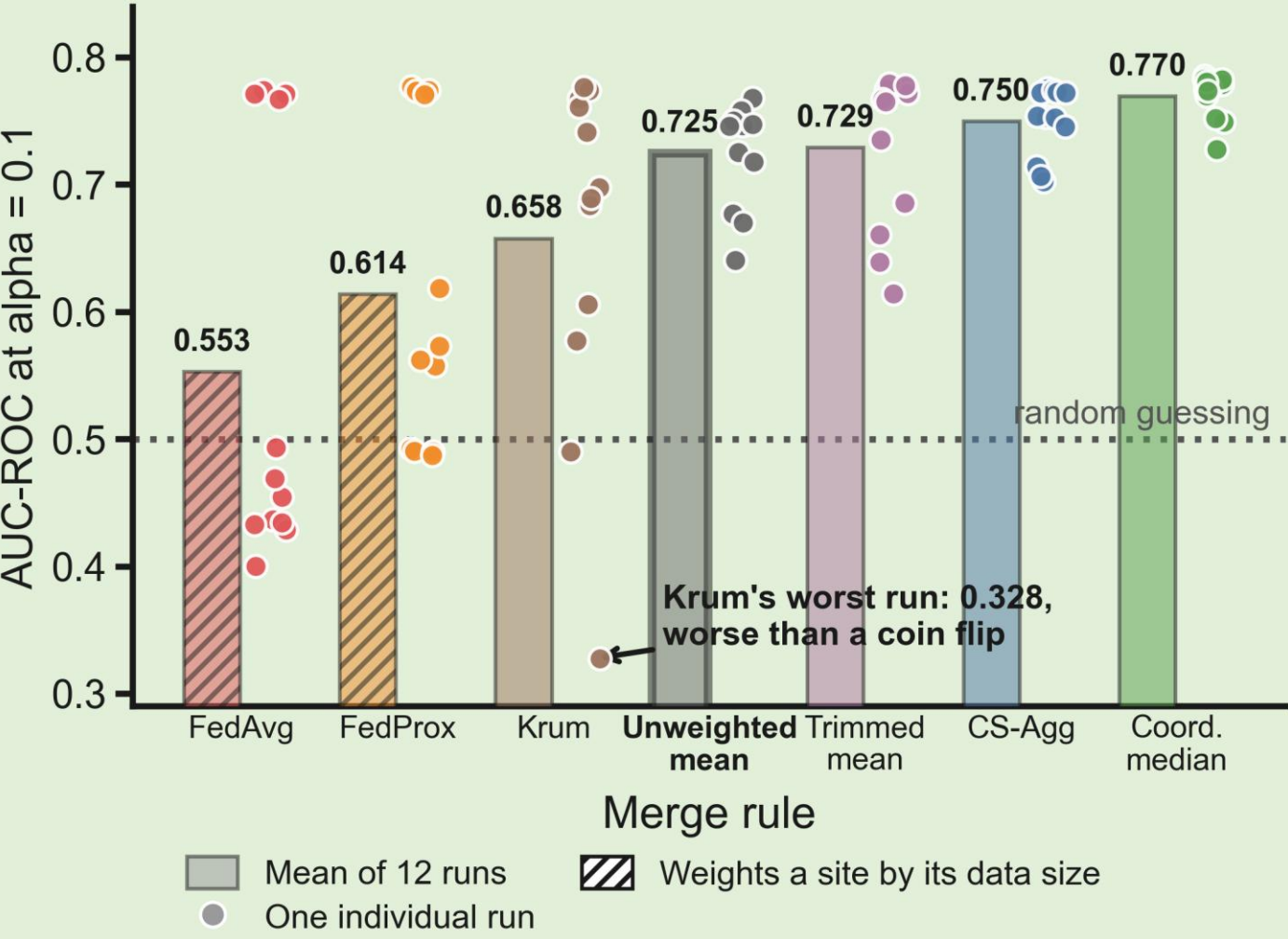


Figure 4. The individual runs behind those means. Krum's average of 0.658 hides a run at 0.328, worse than a coin flip, while the control never falls below 0.64. An average is not evidence a rule is safe.

Results, continued

Lopsided data hurts far more than broken labels

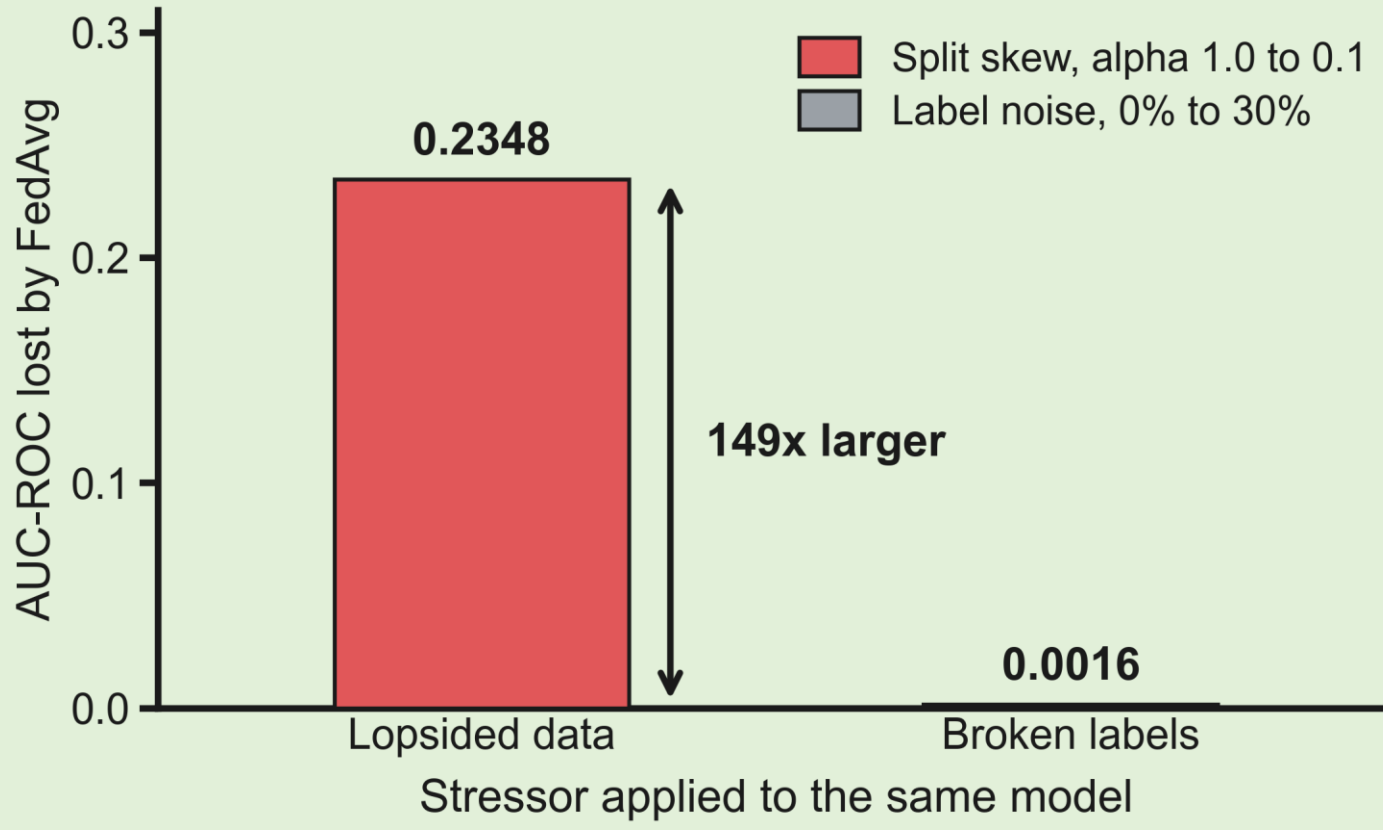


Figure 5. Both stressors, measured on FedAvg. Lopsided data costs 149 times more AUC than broken labels do, so robustness work aimed at noisy labels is aimed at the smaller problem.

- That 149x gap narrows to about 14x when sites track twelve health variables instead of four, but it never closes.

A clean benchmark would have hidden all of this

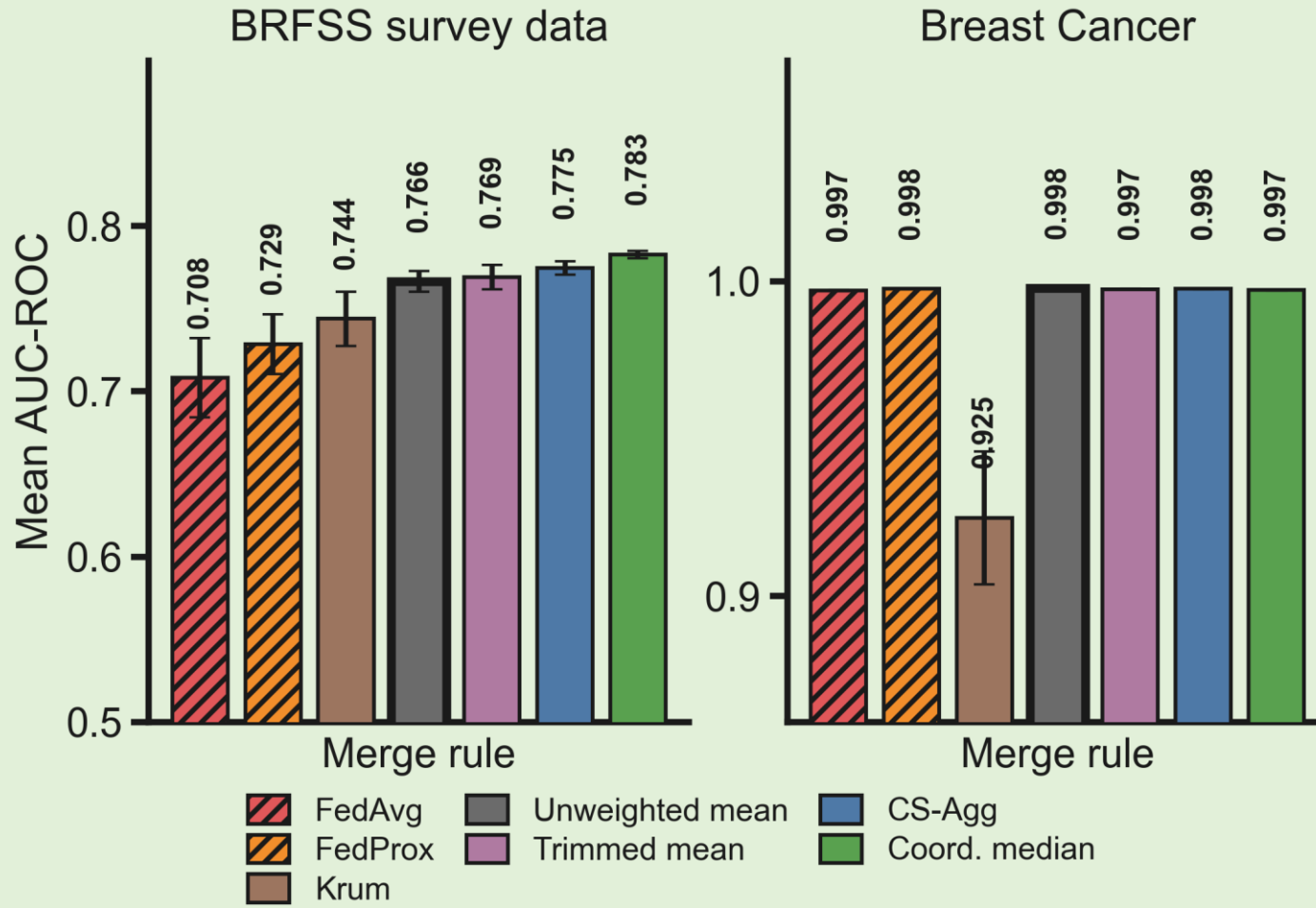


Figure 6. Both datasets, averaged over every split, noise level and seed (252 runs each). On the clean benchmark every rule but Krum reaches 0.997, so a study run only there would have found nothing.

- Messy, self-reported population health data is where these rules actually separate.

Future Directions

- Test adaptive Byzantine attackers, not only the label noise studied here.
- Replace the simulated split with data partitioned across real institutions, and check that the weighting effect holds for deeper models and more sites.

Conclusions

- Most of the advantage robust aggregators hold over FedAvg comes from how they weight sites, not from the robust statistic they are named for.
- Any robustness evaluation run on size-skewed data should include an unweighted-mean control.
- The control costs one line of code, and without it a reported gain cannot be credited to robustness.
- A rule's reputation is not evidence: Krum is widely cited and failed here, while coordinate-wise median is simple and held.

Code & Data

All code, the merged results and a script that re-checks every number on this poster are openly available.

github.com/kunalchevuri/federated-learning-health-robustness



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